

VLA-JEPA - browser rehearsal guide

Noah Sprenger and Fernando Nogueira

The measured two-voice narration runs **7:53**. Noah speaks for 3:56 on scenes 1-8; Fernando speaks for 3:56 on scenes 9-18. Each chapter ends with a short 0.45-second handoff. Pacing stays close to the generated speech rather than stretching to the previous slide budgets.

Browser controls

Arrow keys change scenes. Space plays or pauses the independent motion sequence. Build advances the independent local motion. End or Final view shows the complete scene. Home replays. The chapter rail navigates the whole talk. N toggles notes. Navigation and playback controls appear near the screen edges. Narration continues through the main talk; local motion has a separate clock.

The exported movie is the automatic, narrated version. Browser motion follows the available display refresh rate. The movie export uses constant 60 fps at 1920 x 1080. Hardware and browser conditions can limit interactive frame rate.

The light theme, custom scenes, real example frames and continuous flow curve replace the original full-slide images. The opening and conclusion use a procedural Three.js feature sculpture. The method uses actual SSV2 video frames. Feature ribbons and flow trajectories are explanatory illustrations, not measured model outputs. Shared objects interpolate their positions between method chapters. Benchmark values morph in the same chart stage, and both comparators move together. The robot example displays recorded Cartesian pose commands, normalized per axis. B1-B4 are optional references outside the main narrated talk.

Timing and validation

Scene	Speaker	Start	Length
1	Noah	0:00	17.9s
2	Noah	0:17	30.7s
3	Noah	0:48	40.9s
4	Noah	1:29	44.2s
5	Noah	2:13	21.4s
6	Noah	2:35	22.1s
7	Noah	2:57	23.0s
8	Noah	3:20	36.8s
9	Fernando	3:56	34.8s
10	Fernando	4:31	26.4s
11	Fernando	4:58	30.7s
12	Fernando	5:28	14.0s
13	Fernando	5:42	22.2s
14	Fernando	6:04	19.4s
15	Fernando	6:24	18.6s
16	Fernando	6:43	28.2s
17	Fernando	7:11	24.8s
18	Fernando	7:36	17.7s

Content check

All eighteen narration segments were regenerated from the revised dialogue using Gemini 3.8 TTS. All final scenes and selected intermediate build states were rendered for inspection. These checks establish timing, consistency, and visual fit; they do not measure either presenter's delivery or the audience's understanding. A spoken rehearsal and a classmate comprehension check are still useful.

Analytical flow check

The first half establishes the data gap, related work, predictive supervision, and the information boundary. The second half explains action generation, distinguishes the two training sources, then separates benchmark comparisons from the human-video ablation and the proposed efficiency study. Scenes 4-7 serve different roles: overview, encoding, prediction, and alignment. Implementation settings are kept in notes rather than presented as the audience's main takeaway.

Check audience understanding

After rehearsal, ask a classmate: What does human video supervise? Why are latent tokens not motor commands? How do flow-matching training and inference differ? Which evidence supports robustness, and which counterexample limits the claim? A correct answer should separate future targets from policy inputs and paper results from our proposed quantization comparison.

Course requirements

The narrated movie is under the ten-minute limit and focuses on the method. The deck includes related work and critical result analysis. Upload the final video to the course Teams location and the PDF to the assignment location. Actual timely participation in course discussion cannot be demonstrated by these artifacts.

Sources

VLA-JEPA paper: <https://arxiv.org/abs/2602.10098v1>

Human footage and its three key frames: Something-Something-v2 validation sample 174198, putting jar into box. Retrieved from the public morpheushoc/something-something-v2 archive mirror and checked against validation.json. <https://www.qualcomm.com/developer/software/something-something-v-2-dataset> ; <https://huggingface.co/datasets/morpheushoc/something-something-v2>

Robot video on slide 2: DROID raw episode AUTOLab+0d4edc83+2023-10-27-19h-52m-50s, exterior camera 24400334. Timing restored from recorded control timestamps; muted, cropped, with an end hold. <https://droid-dataset.github.io/visualizer/>

Related work: UniVLA, <https://arxiv.org/abs/2505.06111> ; LAPA, <https://latentactionpretraining.github.io/> .

UniVLA already learns in DINO feature space; the main comparison distinguishes pipelines and the information entering the latent-action pathway.

The precise cue times below are practical speaking checkpoints estimated across each audio segment. They are not word-level forced alignment or experimentally optimal pause timings. The voices are synthetic stock voices, Puck and Charon, generated using google/gemini-3.8-flash-tts.

Scene 1 - Noah

VLA-JEPA

Clock 0:00-0:17; measured allocation 17.9 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:08 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

Hi everyone. We're Noah and Fernando. Today we're looking at VLA JEPA, a method that asks whether watching human videos can help a robot learn to act. We'll walk through how it works, look at the paper's evidence, and finish with the efficiency questions we're exploring in our own project.

Sun et al., VLA-JEPA, arXiv:2602.10098v1 (2026), abstract and introduction.

Scene 2 - Noah

What can video teach a robot?

Clock 0:17-0:48; measured allocation 30.7 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:15 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

Watch these two examples. On the left, a person puts a jar into a box. On the right, a robot puts a block into a bowl. Both show a change in the world. But only the robot demonstration comes with the control commands that produced it. Human videos are easier to collect, yet they don't tell a robot how to move its own joints or gripper. So the question is: can we learn something useful from the visual transition, and connect that knowledge to robot control? That's the gap this paper addresses.

Human footage: Something-Something-v2 validation sample 174198.webm, label putting jar into box, template Putting [something] into [something]. Retrieved from the public morpheushoc/something-something-v2 archive mirror and checked against its validation.json annotation; native 427x240, 12 fps, 5.916 seconds. Cropped to 426x192; output duplicates source frames at 60 fps. Before/during/after stills are from this same clip. This is a verified V2 dataset example, not an identified VLA-JEPA training example. Dataset:

<https://www.qualcomm.com/developer/software/something-something-v-2-dataset> . Mirror:

<https://huggingface.co/datasets/morpheushoc/something-something-v2> . Robot footage: real DROID episode AUTOLab+0d4edc83+2023-10-27-19h-52m-50s, exterior camera 24400334. Cropped; approximate timing restored from the 5.901-second recorded control-timestamp span, with an end hold. Paired trajectory.h5 contains recorded controls. Both videos play once silently, then hold. This is dataset footage, not a VLA-JEPA policy rollout. <https://droid-dataset.github.io/visualizer/>

Scene 3 - Noah

Feature prediction is shared; the pipeline differs

Clock 0:48-1:29; measured allocation 40.9 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:20 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

Before we get into the method, there's an important comparison. LAPA learns discrete latent actions by reconstructing future pixels. UniVLA already works in DINO feature space: it learns discrete codes, then trains a policy to predict them. So feature prediction alone isn't new here. In VLA JEPA, the policy produces continuous latent tokens from the current observation and instruction. Those tokens help predict future state features, and the prediction loss trains the policy representation itself. The difference is how the pieces are connected, and which information goes into them. Robot demonstrations then teach a separate action head to produce control commands.

LAPA (2024), §3; UniVLA (RSS 2025), §III; VLA-JEPA (2026), §3. Different targets and pipelines; feature prediction is not unique to VLA-JEPA.

Scene 4 - Noah

Predict a future state in feature space

Clock 1:29-2:13; measured allocation 44.2 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:22 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

Think about watching someone put a jar into a box. You can anticipate the outcome without drawing every pixel of the next frame. That's the intuition, but let's make the model's prediction precise. The latent action tokens describe a transition. The world predictor takes those tokens, together with encoded state history, and predicts the next state's embedding. A frozen V JEPA two encoder processes the actual future video to give us a target. We compare the two representations. We're predicting future visual features, not an image, and not the robot's joint coordinates. A separate action head handles control. Let's follow those same pieces through the model.

VLA-JEPA, §3.2. Human frames: SSV2 validation example #174198. Diagram features are illustrative, not measured embeddings or explicit robot coordinates.

Scene 5 - Noah

First, encode the observation and instruction

Clock 2:13-2:35; measured allocation 21.4 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:10 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

First, the policy reads the current observation and the instruction. Its vision language backbone is Qwen three V L, with two billion parameters. It produces latent action tokens: an internal representation of the transition the instruction calls for. These tokens will condition prediction and control, but they aren't motor commands yet.

VLA-JEPA, §3.2. Human frames: SSV2 validation example #174198. Diagram features are illustrative, not measured embeddings or explicit robot coordinates.

Scene 6 - Noah

Predict the next state's embedding

Clock 2:35-2:57; measured allocation 22.1 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:11 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

Next, the world predictor combines those tokens with encoded state history. Its output is the next state's embedding. That distinction matters: the tokens represent a transition; the prediction represents the future state. There's no image decoder in this branch. The actual future will supply a training target, rather than an input to the policy.

VLA-JEPA, §3.2. Human frames: SSV2 validation example #174198. Diagram features are illustrative, not measured embeddings or explicit robot coordinates.

Scene 7 - Noah

Compare the prediction with the observed future

Clock 2:57-3:20; measured allocation 23.0 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:11 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

Now we can check the prediction. The frozen encoder processes the actual future frames and produces the target embedding. The alignment loss measures the mismatch. Training updates the prediction path, including the vision language model, while the target encoder stays fixed. So a video teaches the policy which representations are useful for anticipating what happens next.

VLA-JEPA, §3.2. Human frames: SSV2 validation example #174198. Diagram features are illustrative, not measured embeddings or explicit robot coordinates.

Scene 8 - Noah

The predictor sees history, while future states are targets

Clock 3:20-3:56; measured allocation 36.8 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:18 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

There's a boundary we need to keep clear: the policy doesn't get to see the answer. Its latent tokens come from the initial observation and instruction. During training, the world predictor can use earlier states through teacher forcing, but its attention remains causal. It can't use the future state it's meant to predict. Future embeddings belong on the target side of the loss. Otherwise, a good prediction might just be a shortcut, rather than evidence of learning a useful transition. We've now seen how the model learns a representation. Fernando will explain how the action head turns that representation into control.

VLA-JEPA, §3.2. Human frames: SSV2 validation example #174198. Diagram features are illustrative, not measured embeddings or explicit robot coordinates.

Scene 9 - Fernando

Learn the velocity from one training pair

Clock 3:56-4:31; measured allocation 34.8 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:17 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

Thanks, Noah. Let's start with one training example. The blue point is a demonstrated action chunk, and the gray point is sampled noise. We choose a mixing time and construct a point between them. The orange arrow shows the target velocity: the displacement from noise toward the demonstration. The network sees the mixed point, the mixing time, and the latent tokens, and predicts a velocity. We train it to match that target, across many pairs and mixing times. This is a two dimensional sketch; the real points are higher dimensional action chunks.

Paper §3.3, Eqs. 7-8. Mixing time is generation time, not physical robot time. The target velocity is demonstration minus sampled noise; squared-error regression. Own 2D illustration inspired by Jia-Bin Huang and Julia Turc videos; not an empirical trajectory.

Scene 10 - Fernando

Generate actions by following the learned field

Clock 4:31-4:58; measured allocation 26.4 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:13 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

At inference, we don't have a demonstration to aim at. We start with fresh noise and ask the learned velocity field which way to move. We take a small step, ask again at the new position, and repeat. This produces an action chunk conditioned on the latent tokens. The paper uses four integration steps. The curved path is an illustration: a straight training pair doesn't require every generated trajectory to be straight.

Paper §3.3 and Appendix A: four denoising/integration steps, 7-dimensional actions, future action horizon 7. Own illustrative trajectory; sampler type is not asserted. Training and inference paths are different concepts.

Scene 11 - Fernando

Human and robot data provide different supervision

Clock 4:58-5:28; measured allocation 30.7 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:15 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

Now the two training sources fit together. Human video tells us what happened next, so it supports future-state alignment. Robot demonstrations support that objective too, and also provide the action labels for flow matching. The robot loss combines the action objective with a weighted prediction objective. The paper pretrains on roughly two hundred and twenty thousand human videos and seventy six thousand robot trajectories. Human data helps shape the representation; labelled robot data connects it to control.

Paper, equation 9, sections 3.3 and 4.1; SSv2 220K videos, DROID 76K trajectories. Equation 5 is presented as an embedding discrepancy without an explicit norm; this presentation does not invent an L1 or Smooth-L1 choice.

Scene 12 - Fernando

Standard LIBERO: a near tie

Clock 5:28-5:42; measured allocation 14.0 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:07 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

These are the paper's reported results. On standard LIBERO, VLA JEPA and OpenVLA O F T are almost tied: ninety seven point two versus ninety seven point one percent. That alone isn't a strong argument for the method.

Paper, Tables 1 and 3. Differences: $97.2 - 97.1 = 0.1$ percentage points, $79.5 - 69.6 = 9.9$ percentage points. Selected baseline is OpenVLA-OFT; this is not a claim of superiority to every model on every task. Paper reports 50 episodes per task on standard LIBERO and does not provide confidence intervals for these headline averages.

Scene 13 - Fernando

LIBERO-Plus: a larger gain under perturbations

Clock 5:42-6:04; measured allocation 22.2 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:11 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

LIBERO Plus is more revealing because it changes the camera, lighting, and object layout. The authors report seventy nine point five percent for VLA JEPA, versus sixty nine point six for OpenVLA O F T: a nine point nine percentage point gain. That supports robustness under these particular shifts. It doesn't tell us that every failure has been solved.

Paper, Tables 1 and 3. Differences: $97.2 - 97.1 = 0.1$ percentage points, $79.5 - 69.6 = 9.9$ percentage points. Selected baseline is OpenVLA-OFT; this is not a claim of superiority to every model on every task. Paper reports 50 episodes per task on standard LIBERO and does not provide confidence intervals for these headline averages.

Scene 14 - Fernando

Human video improves LIBERO-Plus robustness

Clock 6:04-6:24; measured allocation 19.4 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:09 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

The ablation asks a more direct question: what happens without human video? On LIBERO Plus, the paper reports a drop from seventy nine point five to sixty two point nine percent. That's sixteen point six percentage points within the same method, giving stronger evidence for the contribution of that training source.

Paper, Tables 1-3 and sections 4.4-4.5. Within-method ablation deltas with human video minus without: LIBERO-Plus +16.6 pp; LIBERO +1.1 pp; SimplerEnv Google -13.2 pp. Real-world repeated grasping is an author-reported qualitative observation, with a proposed attribution to human videos, not an isolated causal proof.

Scene 15 - Fernando

Human video does not improve every benchmark

Clock 6:24-6:43; measured allocation 18.6 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:09 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

But the benefit isn't universal. On the Google robot tasks in SimplierEnv, adding human video reduces reported success from seventy eight point four to sixty five point two percent. The strongest evidence here is for LIBERO Plus robustness, rather than improvement on every benchmark.

Paper, Tables 1-3 and sections 4.4-4.5. Within-method ablation deltas with human video minus without: LIBERO-Plus +16.6 pp; LIBERO +1.1 pp; SimplierEnv Google -13.2 pp. Real-world repeated grasping is an author-reported qualitative observation, with a proposed attribution to human videos, not an isolated causal proof.

Scene 16 - Fernando

Our proposal: evaluate efficiency without assuming success

Clock 6:43-7:11; measured allocation 28.2 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:14 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

That brings us to our project proposal. We're asking whether this approach can be made more practical on a single GPU. We plan to compare reduced precision against the same baseline, measuring memory, latency, and task success together. We're also exploring a smaller vision language backbone while keeping the predictive and action components. So far, our work is on the implementation and evaluation setup. We're not announcing project performance results here, and we don't yet know which approach will offer the best trade-off.

Presenters' project proposal and work in progress. No project performance results, speedups, success retention or memory reductions are announced. Planned comparisons require matched evaluation; outcomes remain open.

Scene 17 - Fernando

Deployment gaps remain beyond headline averages

Clock 7:11-7:36; measured allocation 24.8 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:12 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

The paper also leaves open questions. Under sensor noise, VLA JEPA trails pi zero in this comparison. Its real robot study uses only ten trials per task, with some wrong object selections. And training used eight A one hundred GPUs. Our proposed single GPU study concerns efficiency and deployment; it doesn't establish that reproducing the paper's training is cheap.

Sun et al., Table 3, §4.1, §4.4 and Appendix B. Ten trials per task describes the authors' real-world study; eight A100 GPUs describes their training. Our single-GPU work is a proposed efficiency comparison, with no performance claims.

Scene 18 - Fernando

Predictive supervision strengthens the policy

Clock 7:36-7:53; measured allocation 17.7 seconds.

0:00 - Begin as a connected explanation; avoid a dramatic reset.

0:08 - Place gentle emphasis on the main distinction; leave a short breath between ideas.

Dialogue

Our takeaway is that predicting future features can help train a robot policy, with the clearest evidence here coming from robustness under perturbations. For our project, the question remains open: can we reduce the resource cost while keeping useful control? That's what we want to evaluate. Thank you.

Paper, sections 3 and 4.5. Suggested discussion experiment: controlled appearance changes versus changed dynamics, under matched data and compute. This is our proposed test, not a result in the paper.

Implementation reference

The reported world predictor has 12 layers, 2 camera views, and an 8-frame future horizon. The action head uses 7-dimensional actions with four denoising/integration steps. These are reproduction details; the main scenes focus on what information flows through the model and what training objective teaches it.